United States Patent	12398550
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wedi; Stephan

Adapter for a drain device

Abstract

An adapter for a drain device, for installation in a floor panel, and having a receptacle with an upper inlet opening through which waste water can flow into the drain device, and parts which can be connected in a fluidic manner to a drain pipe and the receptacle, so that waste water which has passed through the inlet opening can enter the drain pipe. To improve the sealing function, the adapter has a base part having an inner thread, a first coupling point to the drain pipe, a second coupling point to the receptacle and which couples the receptacle to the drain pipe, a sealing ring that can be installed in the base part to seal the adapter from the drain pipe, and a compression ring with an outer thread compatible to the inner thread of the base part for exerting a compressive force on the sealing ring.

Inventors:	**Wedi; Stephan (Emsdetten, DE)**
Applicant:	**wedi GmbH** (Emsdetten, DE)
Family ID:	**1000008779849**
Assignee:	**WEDI GmbH (Emsdetten, DE)**
Appl. No.:	**17/771052**
Filed (or PCT Filed):	**March 17, 2022**
PCT No.:	**PCT/EP2022/056935**
PCT Pub. No.:	**WO2023/174540**
PCT Pub. Date:	September 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240151022 A1	May. 09, 2024

Publication Classification

Int. Cl.: **E03F5/04** (20060101); **F16J15/10** (20060101)

U.S. Cl.:

CPC **E03F5/0408** (20130101); **F16J15/106** (20130101);

Field of Classification Search

CPC: E03F (5/0408); E03F (4/0407); F16J (15/106)

USPC: 4/679

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3896511	12/1974	Cuschera	4/288	E03C 1/22
4622703	12/1985	Cuschera	277/606	E03C 1/22
11268271	12/2021	Say	N/A	E03F 5/0411
11284751	12/2021	Droessler	N/A	A47K 3/405
2015/0275497	12/2014	Nyce	52/302.1	E03F 5/0407
2018/0279840	12/2017	Ruiz Perez	N/A	E03F 5/0409
2022/0195712	12/2021	Wedi	N/A	F16J 15/106

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102019110322	12/2019	DE	N/A

Primary Examiner: Le; Huyen D

Attorney, Agent or Firm: Smith Tempel Blaha LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This patent application claims the benefit of and priority on International Application No. PCT/EP2022/056935 having an international filing date of 17 Mar. 2022.

BACKGROUND OF THE INVENTION

Technical Field

(2) The invention relates to an adapter for a drain device that is provided for installation in an opening of a floor panel, preferably a shower floor panel. Such adapters are used to connect a drain body installed in a floor surface with a connecting piping system.

Prior Art

(3) Known from DE 10 2019 110 322 A1 is an adapter that can be mounted on a drain body and which produces a connection between said drain body and a piping system. Used water that initially collects in the drain body is thus directed through the adapter into the piping system. The

adapter is sealed from the piping by means of a special seal which has an annular outer contour that can inserted into a contour provided on the adapter. In order to couple the adapter to the piping system, the seal can first be fitted on the drain pipe. Then the adapter is attached.
(4) The seal has an upper, inward-facing collar that protrudes over the pipe end of the drain pipe from the outside to the inside when mounted. There is also a collar on the underside of the seal into which a pipe stub protruding from the drain body can be inserted. Furthermore, the seal comprises flexible sealing lips that fit snugly against the outer wall of the drain pipe when the adaptor is pushed onto it. The sealing lips exert a relatively low contact pressure on the drain pipe on the one hand and also seal it on the other. The pressure exerted by the seal on the drain pipe depends on the seal material and the tightness of the fit: The tighter the fit and the harder the seal material, the higher the pressure exerted by the seal lips on the drain pipe, but at the same time the more difficult it is to mount the adapter on the drain pipe.

BRIEF SUMMARY OF THE INVENTION

(5) The object of the invention is to overcome the described disadvantages and to propose an adapter which is easy to install and provides a fixed, but nevertheless easily detachable connection between the drain pipe and the drain body
(6) This object is achieved by an adapter for a drain device that is provided for installation in an opening of a floor panel, preferably a shower floor panel, and wherein the drain device comprises a receptacle for the installation in the floor panel with an upper inlet opening through which waste water can flow into the drain device, and parts which can be preassembled in the floor area and which can be connected in a fluidic manner to a drain pipe and the receptacle, so that waste water which has passed through the inlet opening can enter the drain pipe, characterized in that the adapter comprises the following components: a base part having an internal thread, wherein the base part comprises a first coupling point to the drain pipe and a second coupling point to the receptacle and couples the receptacle to the drain pipe in the assembled state, a sealing ring that can be installed in the base part to seal the adapter from the drain pipe, and a compression ring with a outer thread compatible to the inner thread of the base part for exerting a compressive force on the sealing ring, so that the sealing ring can be pressed against the drain pipe by means of the compression ring and the first coupling point comprises a press-fit connection.
(7) The adapter according to the invention is provided for a drain device, which preferably may be a shower floor drain device. The drain device comprises a receptacle that can be installed in a floor panel and a drain pipe that can be flow-connected to the receptacle. The receptacle is thus understood to be a component that can be installed in the base panel and which forms a kind of water inlet housing.
(8) The adapter itself comprises the following components: a base part which has a first coupling point to the drain pipe and a second coupling point to the receptacle and couples the receptacle to the drain pipe in the assembled state, a sealing ring that can be installed in the base part to seal the adapter from the drain pipe, and a compression ring for exerting a compressive force on the sealing ring
(9) The sealing ring comprises a deformable, flexible material, for example plastic or rubber. Preferably, a receptacle compatible with the sealing ring, for example a groove, is provided in the base element. The sealing ring can be easily inserted into such a groove and securely fixed at the intended installation location. As an alternative or in addition to a groove, the base element can also have a lower collar, on whose contour the sealing ring inserted into the base element can be supported.
(10) A clearance fit is provided between the sealing ring and the drain pipe in the unloaded state, which is dimensioned in such a way that the adapter with the installed sealing ring can be easily fitted onto the drain pipe. In this state, the sealing ring does not yet exert a sealing function. Rather, the sealing function is only attained by the sealing ring in the adapter being deformed by means of a force exerted on the sealing ring in such a way that it not only closes a gap formed between the

drain pipe and the adapter, but is also deformed or crimped in this gap with a high compressive force.
(11) The force exerted on the sealing ring is applied by screwing the compression ring into the base element. The compression ring is preferably not screwed into the base element manually, but with a suitable tool, since a tool can exert a higher torque in the compression ring and thus also a higher pressure force on the sealing ring. As the compression ring is screwed in, its underside comes closer and closer to the sealing ring until the compression ring and the sealing ring finally make contact with each other. If the screwing-in process is continued, the compression ring presses on the sealing ring, deforms it and creates the desired compression connection between the drain pipe and the adapter. As the gap between the drain pipe and the adapter is completely filled by the deformed sealing ring during this pressing process, the drain pipe and adapter are also reliably sealed.
(12) Preferably, the compression ring is not screwed into the base element by means of an external tool, but by means of an insert integrated into the adapter. This insert may be a part which can be fitted into the base element above the compression ring and which is provided for tensioning the sealing ring fitted into the base element. The compression ring and the insert preferably have connecting elements that are compatible with each other to create a form-fit connection. This makes it very easy to apply the torque required to screw in the compression ring first to the insert and then transfer it from the insert to the compression ring. The sealing ring can thus be pressed against the drain pipe particularly easily and with high pressure force by means of the combination of compression ring and insert. The compatible connecting elements can be, for example, a type of toothing.
(13) A tight clearance fit is preferably provided between the compression ring and the insert, so that the insert can be easily separated from the compression ring on the one hand, but on the other hand a torque acting on the insert can also be securely transmitted to the compression ring.
(14) In one preferred embodiment of the invention, a spacer is provided between the sealing ring and the compression ring. A spacer is a tube-like part that can be inserted into the base part of the adapter and forms a kind of spacer between the compression ring and the sealing ring. In the adapter, the spacer is located between the sealing ring and the compression ring when mounted. The spacer is pressed onto the sealing ring by the screwing-in movement of the compression ring. The underside of the compression ring, which rotates as it is screwed in, does not rub directly against the flexible sealing ring, but against the top side of the spacer. Preferably, the spacer itself does not rotate or rotates only insignificantly despite the rotational movement of the compression ring. The spacer is thus moved linearly in the direction of the compression ring.
(15) The top side of the spacer is designed in such a way that it is not damaged by the aforementioned rotational friction of the compression ring. Put simply, the linear-moving spacer merely compresses the sealing ring, but does not rub along the sealing ring.
(16) In a preferred embodiment, the spacer comprises a seal for additional sealing of the adapter against the drain pipe. Sealing is thus achieved on the one hand by the compressible sealing ring and on the other hand by the seal provided on the spacer. This seal provided on the spacer has an inner diameter that is slightly smaller than the outer diameter of the drain pipe. The seal thus forms a guide when the adapter is fitted onto the drain pipe. In addition, the seal closes any gap remaining between the drain pipe and the housing when the drain pipe is inserted flush. In addition to the improved appearance, no waste water can penetrate and remain there, so that unpleasant odors are avoided.
(17) Preferably, the seal fitted in the spacer is glued in place. It can therefore neither be forgotten nor incorrectly mounted during assembly.
(18) The spacer can have at least one nominal separation point at which it can be divided. Preferably, the tubular spacer can comprise two nominal separation points by means of which it can be divided into two shell-like halves. Such a division enables the subsequent installation of a spacer

in an adapter even if a receptacle located above the adapter in the base panel or a component of the drain device located on the receptacle, for example a narrow channel drain, only has a passage to the drain pipe that is smaller than the diameter of the spacer.
(19) Although the spacer is usually inserted into the adapter before the adapter is assembled, it can happen that the installation of the spacer is forgotten during assembly. In order to allow the spacer to be pushed through an opening in the receptacle that is smaller than the diameter of the spacer, the spacer is preferably divided into two parts at the intended nominal separation points. The two half-shells created in this way can be inserted one after the other through the opening of the inlet housing and then positioned in the adapter above the sealing ring at the intended location.
(20) In cases where the opening of the inlet housing is only minimally smaller than the spacer, it may also be sufficient to separate the spacer only at a nominal separation point. A spacer split in this way can be twisted in a spiral movement and pushed in one piece from above through the opening of the inlet housing into the adapter.
(21) The first coupling point, which connects the adapter to the drain pipe, is formed as already explained by the compression ring being screwed into the receptacle and consequently compressing the sealing ring and pressing it between the receptacle and the drain pipe. The compression ring and receptacle thus each comprise a thread. In order to position these two threads in the correct position in relation to each other when screwing them in and to prevent the part provided with the external thread from being placed obliquely in relation to the part provided with the internal thread, a bayonet lock can be provided which is integrated into the threaded connection—and thus also into the coupling point.
(22) A bayonet lock is a mechanical connection of two cylindrical parts in their longitudinal axis that can be quickly made and released. The parts are connected and separated by inserting them into each other and turning them in opposite directions. In the case of a bayonet lock integrated in a thread, the two components are first turned against the direction of rotation of the thread with little pressure to ensure correct positioning. In a predefined position of the two parts in relation to each other, the thread is interrupted and the part provided with the external thread slides over the part provided with the internal thread in the direction of the longitudinal axis of the thread. In simple terms, the two threads snap into each other and are thereby positioned in the correct position in relation to each other. The direction of rotation can then be changed and the internal thread can be screwed into the external thread.
(23) In a preferred embodiment, the insert may comprise openings for collecting impurities present in the waste water, for example hair, and thus have a sieve-like effect. The insert thus has a dual function: on the one hand, it serves to transmit a torque to the press ring and to create the press connection between the adapter and the drain pipe; on the other hand, it serves to prevent foreign bodies from entering the drain pipe and thus to reduce the risk of pipe blockages.
(24) Preferably, the insert comprises a receptacle for a tool, in particular a screwdriver, so that the insert can be inserted or removed from the adapter by means of a tool. This receptacle can be, for example, a slot, a cross-slot or a Torx connection. The receptacle is preferably mounted in the center axis of the insert part so that a torque introduced into the insert part with the tool via the receptacle can be transmitted evenly to the compression ring coupled to the insert part. It is therefore particularly advantageous that no special tools are required to screw in the insert.
(25) The second coupling point, which connects the adapter to the receptacle and thus to the water inlet housing, is preferably formed by a threaded connection. For this purpose, a thread can be provided on the adapter which is compatible with a thread attached to the receptacle. Preferably, this screw connection comprises a seal which is compressed when the adapter is screwed into the receptacle and seals the transition area from the adapter to the receptacle.
(26) Analogous to the first coupling point, the second coupling point can also comprise a bayonet lock. For this purpose, the thread of the adapter can be formed as a first part of a bayonet lock, which is compatible with a second part of the bayonet lock attached to the receptacle. This results

in the advantages already described in connection with the bayonet lock of the first coupling point of a simple and coaxial positioning of the components to each other.
(27) The second coupling point may further comprise, on one of the two components, namely the adapter or the receptacle, a catch which is engageable in a recess of the other component which is compatible therewith. When the threaded connection reaches a predefined insertion depth, the catch and recess arrive in a mutually compatible position. The catch, which is preferably elastic and in a pre-tensioned position when the threads are screwed in, relaxes in this position and engages in the recess. This fixes the adapter, which can be screwed into the receptacle, and it can only be unscrewed from the receptacle when the catch is deliberately pushed out of the recess, usually with the aid of a lever-like tool. Only then is it possible to remove the adapter from the receptacle. The catch thus forms a safeguard against unintentional unscrewing. Such unintentional unscrewing can occur if other parts of the drain device, which also comprise a threaded connection, are to be loosened and the torque applied is wholly or partially not transferred to the thread to be loosened, but to the threaded connection of the receptacle and adapter.
(28) In a preferred embodiment, it may be provided that the first coupling point for connecting the adapter to the drain pipe and the second coupling point for connecting the adapter to the receptacle each comprise a thread oriented in opposite directions. This not only prevents the other coupling point from being loosened when one coupling point is tightened, but on the contrary, one coupling point is additionally tightened when a torque is applied to the other coupling point.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Further measures improving the invention are illustrated in more detail below with the description of preferred embodiments of the invention with reference to the figures, which show:
(2) FIG. **1** is a drain device installed in a floor panel in a perspective sectional view;
(3) FIG. **2** is the floor panel with drainage device according to FIG. **1** in a sectional side view;
(4) FIG. **3** is a floor panel and a drain device to be installed in the floor panel in an exploded view;
(5) FIG. **4** are some components of the drainage device which together form an adjusting device in a cut side view;
(6) FIG. **5** is a detail enlargement from FIG. **4**;
(7) FIG. **6** are the components according to FIG. **4** in a sectional perspective view;
(8) FIG. **7** is a detail enlargement from FIG. **6**;
(9) FIG. **8** is a fixing element in the form of a snap ring in a top view;
(10) FIG. **9** is the fixing element according to FIG. **8** in a view from below;
(11) FIG. **10** is an extension element for a cover in a perspective view;
(12) FIG. **11** is a retainer for the extension element shown in FIG. **10** in a perspective view;
(13) FIG. **12** is a receptacle built into a floor panel in a perspective view;
(14) FIG. **13** is a perspective view of the receptacle according to FIG. **12** with the retainer installed according to FIG. **11**;
(15) FIG. **14** is a perspective view of the receptacle with retainer and snap ring in the pre-assembled state;
(16) FIG. **15** is an adapter coupled to a receptacle in a perspective view;
(17) FIG. **16** is a perspective sectional view of the adapter coupled to the receptacle according to FIG. **15**;
(18) FIG. **17** is the base part of the adapter in a perspective view;
(19) FIG. **18** is the base part of the adapter according to FIG. **17** in a perspective sectional view;
(20) FIG. **19** is the spacer of the adapter in a perspective view;
(21) FIG. **20** is the spacer of the adapter according to FIG. **19** in a perspective sectional view;

(22) FIG. **21** is the sealing ring of the adapter in a perspective view;
(23) FIG. **22** is the compression ring of the adapter in a perspective view; and
(24) FIG. **23** is an insertable part that can be inserted into an adapter.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(25) In the following figures, identical or similar elements can be provided with identical or similar reference numbers. Furthermore, the figures of the drawings contain numerous features in combination in the description as well as in the claims. It is clear to a person skilled in the art that these features can also be considered individually or that they can be combined to form further combinations not described in detail here. The invention expressly extends also to such embodiments which are not given by combinations of features from explicit back references in the claims, whereby the disclosed features of the invention can be combined with each other in any way, inasmuch as this is technically reasonable. The exemplary embodiments shown in the figures thus have only a descriptive character and are not intended to limit the invention in any way.
(26) Terms such as “upper”, “lower”, “below”, “left”, or “right” refer to the positioning of the components of the drain device as represented in the drawings which corresponds to the arrangement in the operating state.
(27) FIG. **1** shows a drain device **100** installed in a floor panel **10** in a perspective sectional view. The floor panel **10** in the shown exemplary embodiment is a shower floor panel **11**, which has a slope directed towards an inlet opening **13**. Waste water entering through the inlet opening **13** is fed to a drain pipe system **16** connected to the drain device **100**.
(28) In the illustrated exemplary embodiment, the drain device **100** is installed in a sanitary room and is fitted in an opening **12** of the shower floor panel **11**. The drain device **100** shown does not comprise an anti-odor trap integrated into the drain cup. In other exemplary embodiments, it may also be provided that the drain device itself comprises an anti-odor trap.
(29) FIG. **2** shows the floor panel **11** with the drainage device **100** in a side sectional view. An adapter **200** is fitted onto the drain pipe **16** and fixed at a first coupling point **41** by means of a press-fit connection **56**. At a second coupling point **42**, the adapter **200** is screwed to a receptacle **40** by means of a screw connection **62**. A bayonet catch **63** is integrated into the screw connection **62**. The bayonet catch **63** facilitates the alignment of the components to be screwed together.
(30) Further details of the adapter **200** are described in more detail below in relation to FIGS. **15** to **23**.
(31) Together with a retainer **25** and a fixing element **30**, which in the present case is a snap ring **33**, the receptacle **40** forms an adjusting device **150**. The retainer **25** of the adjusting device **150** is connected to an extension element **15** via a thread. An intermediate support **66** may be arranged on the extension element **15**, on which a cover **14** is provided for closing the inlet opening **13**. The cover **14** can be removed to carry out cleaning and maintenance work. In the fully assembled state, the surface of the cover **14** is preferably aligned flush with the surface of a covering applied to the shower floor **11**, which may be floor tiles, for example.
(32) The flush fit of cover **14** to the surface of the shower floor covering is adjusted during installation of the shower floor **11**. In FIG. **2**, it can be seen that the extension element **15** clearly projects above the surface of the shower floor **11**. This is due to the fact that the height of the extension element **15** in the initial stage is initially very generously dimensioned to also allow for the use of thick floor coverings. In order to roughly adjust the height of the extension element **15** to the height of a selected floor covering, the extension element **15** is therefore first shortened on its underside so that it is roughly adjusted to the height of the selected floor covering. The fine height adjustment, accurate to the millimeter or even to the tenth of a millimeter, is carried out by means of the threaded connection of extension element **15** and retainer **25** after the floor covering has been applied.
(33) The adjusting device **150** provides, on the one hand, the described height adjustment and, on the other hand, a horizontal adjustment of the cover **14** for uniform alignment with respect to the

joints surrounding the cover **14**.
(34) FIG. **3** shows an exploded view of the components of the drain device **100**. The cover **14** can preferably be placed in a form-fitting manner on an intermediate support **66** provided in the illustrated embodiment. The intermediate support **66** may also have positive locking contours and is thus able to be placed on the extension element **15** in a form-fitting manner.
(35) The extension element **15** has an external thread **17** that can be screwed into an internal thread **29** of the retainer **25**.
(36) The retainer **25** is held in a pre-assembled state Z**1** in a housing-like structure and is movable in this housing-like structure. The housing-like structure is located in an area between the fixing element **30**—in the present case: the snap ring **33**—and the receptacle **40**.
(37) In the illustrated exemplary embodiment, the receptacle **40** can be inserted into the opening **12** of the floor panel **10** from above. To assemble the drain device **100**, the retainer **25**, the snap ring **33**, the extension element **15**, the intermediate support **66**, and the cover **14** are also inserted into the drain device **100** from above and can be easily dismantled if necessary, for example for cleaning work.
(38) In contrast, a base part **20** provided on the adapter **200** is first attached from below to the receptacle **40** already inserted in the floor panel **10**. Next, the floor panel **10** with the inserted receptacle **40** and the base part **20** of the adapter **200** attached to the underside of the receptacle **40** can be fitted onto the drain pipe **16**. The base part **20** is fixed to the drain pipe **16** by means of the press-fit connection **56** formed by the sealing ring **43**, the spacer **49**, the compression ring **44**, and the insert **45**. The exact functionality is explained in more detail below.
(39) FIGS. **5** to **8** illustrate the construction and operation of the adjusting device **150**, with FIG. **5** showing an enlarged section A**1** from FIG. **4** and FIG. **7** showing an enlarged section A**2** from FIG. **6**.
(40) The adjusting device **150** is shown in the pre-assembled state Z**1**. In this pre-assembled state Z**1**, the extension element **15** is screwed with its external thread **17** into the internal thread **29** of the retainer **25**.
(41) The retainer **25** has a circumferential flange **67** with an outer contour **26**. In the pre-assembled state Z**1**, the flange **67** rests on a bearing surface **21** (cf. FIG. **12**) provided in the receptacle **40**. Above the flange **67**, a snap ring **33** provided with projections **34** is engaged in recesses **23** of the receptacle **40** compatible with the projections **34**. The projections **34** and the recesses **23** form a snap-fit connection **18** between the snap ring **33** and the receptacle **40**.
(42) Snap ring **33** and receptacle **40** thus form a housing-like structure that has approximately the shape of a horizontal U or a circumferential groove that is open towards the center axis. This groove surrounds the flange of the retainer **25** in the pre-assembled state Z**1**. The retainer **25**—and thus also the extension element **15**—are movable in horizontal direction RH and vertical direction RV within the groove. This mobility is indicated by a horizontal play SH, which in the illustrated embodiment is 5 mm, and a vertical play SV, which is less than 1 mm and serves to prevent jamming of the flange **67** between the snap ring **33** and the receptacle **40**.
(43) The horizontal and vertical play is maintained until a floor covering is applied to the shower floor panel **11**. This means: In the course of the installation activities, the floor panel **10** and the drain device **100** are installed first. After the floor panel is in place and connected to the pipe **16**, a floor covering (not shown in the drawings) is applied to the floor panel. After the floor covering has been applied, the height of the cover **14** placed on the extension element **15** can be adjusted exactly to the height of the floor covering by screwing the extension element **15** into the retainer **25**. Furthermore, the horizontal positioning of the cover **14** can also be finely adjusted, since the adjusting device **150** is still in the pre-assembled state Z**1** at this point.
(44) After the position of the cover **14** has been finely adjusted in the vertical and horizontal directions RV, RH, the remaining open joints around the cover **14** are then finally grouted with a suitable mortar. This transfers the adjusting device **150** from the pre-assembled state Z**1** to a fully

assembled state. In the fully assembled state, the adjusting device **150** is thus fixed and only the cover **14** can still be removed from its holder.

(45) FIGS. **8** and **9** show a fixing element **30** in the form of a snap ring **33** in a perspective view from diagonally above (FIG. **8**) and diagonally below (FIG. **9**). In the exemplary embodiment shown, the snap ring **33** is made of a flexible plastic and includes four projections **34** that are insertable into the compatible recesses **23** of the receptacle **40** to form a snap-fit connection **18**. Furthermore, the snap ring **33** has some continuous openings **19** which, in the fully assembled state, serve to drain off seepage water entering the drain.

(46) FIG. **9** also shows a projection **36** which is provided on the underside **32** of the snap ring **33** and which, in the pre-assembled state Z**1** of the adjusting device **150**, is arranged in a recess **35** provided in the retainer **25**, where it limits the horizontal play SH of the retainer **25**.

(47) FIG. **10** shows the extension element **15**, which has a tube-like contour with an external thread **17** and a flange **69** adjoining the tube-like contour. Openings **68** are made in the tube-like contour, which serve to drain off seepage water in the same way as the openings **19** in the snap ring **33**. The tube-like contour can be shortened by a piece of length **70** at the installation site, for example with a cutter knife, in order to adapt it to the thickness of the floor covering to be applied.

(48) FIG. **11** shows the retainer **25**, which also has openings **24** for the drainage of seepage water. The retainer **25** has an internal thread **29** into which the extension element **15** can be screwed. Furthermore, the retainer **25** comprises the aforementioned recess **35** for limiting its movement in horizontal direction RH. The retainer **25** further comprises a flange **67** with a circular outer contour **26**. The outer contour **26** has a diameter D**1** which is smaller than a diameter D**2** of the receptacle **40**. The retainer **25** can thus be inserted into the receptacle **40**.

(49) FIGS. **12** to **14** show the shower floor panel **11** with inserted receptacle **40** in various stages of pre-assembly.

(50) In FIG. **12**, only the receptacle **40** is inserted into the opening **12** of the floor panel **10**. From the lower side of the floor panel **10**, the base part **20** of the adapter **200** is screwed to the receptacle **40**. The receptacle **40** has an inner contour **22** with a bearing surface **21** and recesses **23**.

(51) FIG. **13** shows the receptacle **40** with the retainer **25** inserted. The retainer **25** rests with its flange **67** on the bearing surface **21** of the receptacle **40** and is movable in the horizontal direction RH within the horizontal play SH.

(52) FIG. **14** shows the receptacle **40** with the snap ring **33** mounted. The snap ring **33** is detachably connected to the receptacle **40** via the snap-fit connection **18**. This installation situation represents the pre-assembled state Z**1**, in which the retainer **25** (not shown in FIG. **14**) is movable in the housing-like structure formed by snap ring **33** and receptacle **40** and can thus be adjusted as required.

(53) FIGS. **15** to **23** show the adapter **200** already mentioned above and its components. The adapter **200** serves to couple a water inlet inserted into the floor panel **10** to the drain pipe **16** in a watertight manner. The adapter **200** thus has two coupling points, namely the first coupling point **41**, at which it is coupled to the drain pipe **16**, and the second coupling point **42**, at which it is coupled to the water inlet or to the receptacle **40** integrated in the water inlet (cf. FIG. **2**).

(54) In the illustrated exemplary embodiment, the drain device **100** comprises both an adjusting device **150** and an adapter **200**. Further above, the details of the receptacle **40** with specific features for integrating an adjusting device **150** have already been described. However, with regard to the adjusting device **150**, it is irrelevant whether a drain device on which the adjusting device **150** is to be used comprises an adapter **200**. Analogously, with regard to the adapter **200**, it is also irrelevant whether the drain device **100** comprises an adjusting device **150**. Rather, both assemblies can be universally integrated into a drainage device either alone or together.

(55) FIGS. **15** and **16** show the drain device **100** which, on the one hand, comprises the adapter **200** and, on the other hand, is also intended to integrate an adjusting device **150**. This is evident from the fact that the receptacle **40** shown in FIGS. **15** and **16** also comprises features of the adjusting

device **150**.

(56) As already mentioned, the adapter **200** comprises two coupling points **41** and **42**. The function of the adapter **200** in association with the first coupling point **41** is to produce a watertight connection between the base part **20** and the drain pipe **16**. This fixed connection is the press-fit connection **56** already mentioned in reference to FIG. **2**. In a simple embodiment, it can be provided that the press-fit connection **56** with respect to the drain pipe **16** is realized solely by an adapter with the components base part **20**, sealing ring **43** and compression ring **44**. In the exemplary embodiment shown, however, the adapter **200** also comprises the spacer **49** and the insert **45**.

(57) The press-fit connection **56** is achieved by a screw connection **38**. The screw connection **38** is formed by an internal thread **54** arranged on the base part **20** and an external thread **55** arranged on the compression ring **44**. The compression ring **44** can thus be screwed into the base part **20**. The base part **20** has a lower collar **48** on its underside with an inner diameter D**3** that is smaller than an outer diameter D**4** provided on the sealing ring **43**. With a base part **20** fitted onto the drain pipe **16** with the sealing ring **43** arranged on the collar, the sealing ring **43** cannot therefore move downwards out of the base part **20**. Rather, it is crimped when the compression ring **44** is screwed deep enough into the base part **20**. By crimping the sealing ring **43**, the aforementioned press-fit connection **56** can be realized.

(58) In the exemplary embodiment shown in FIGS. **15** and **16**, the adapter **200** further comprises the spacer **49** and the insert **45**. However, these parts are not absolutely necessary for making the press-fit connection **56**. Rather, the press-fit connection **56** can also be produced by pressing the compression ring **44** directly against the sealing ring **43** without the spacer **49** acting as a spacer. For this purpose, the compression ring could, for example, have a cylindrical contour similar to the contour of the spacer, so that the compression ring acts simultaneously as a spacer and as a pressing means.

(59) The inclusion of an insert **45** in the adapter **200** is also not mandatory. The insert **45** is provided for screwing the compression ring **44** into the base part **20**. This insert part **45** comprises connecting elements **46**′ for producing a positive connection **47** with compatible connecting elements **46** on the compression ring **44**. In addition, the insert **45** comprises a receptacle **59**, which in the exemplary embodiment shown is a slot for attaching a screwdriver. By means of the screwdriver, the insert **45** and the compression ring **44** coupled to the insert can be easily screwed into the base part **20**.

(60) The insert **45** thus forms an insertion aid. The compression ring **44** can also be screwed into the base part **20** without the insert **45**. If an adapter **200** is to be used without an insert **45**, the compression ring **44** could, for example, comprise a shoulder onto which a screwing-in tool can be placed directly, so that a sufficiently strong press-fit connection can be made without using an insert **45**.

(61) FIGS. **17** and **18** show the base part **20** of the adapter **200**. At the lower end is the lower collar **48**, which has the inner diameter D**3**. In the assembled state, the sealing ring **43** is arranged on the lower collar **48**. Furthermore, it can be seen that the base part **20** has an upper collar **50** with a U-shaped recess **65** into which a seal **39** is glued.

(62) The base part **20** has an external thread **60** arranged on the upper collar **50**, which together with an internal thread **61** arranged on the receptacle **40** forms a screw connection **62** and thus the second coupling point **42**. In order to seal this coupling point **42** in a watertight manner, the circular seal **39** glued into the base part **20** is provided in the exemplary embodiment shown. When the external thread **60** is screwed into the internal thread **61**, i.e., when the base part **20** is screwed into the receptacle **40**, the seal **39** is pressed against a circular contour provided in the receptacle **40**. As the depth of engagement increases, the seal **39** is compressed to such an extent that water draining through the drain device **100** cannot exit the drain device **100** at the second coupling point **42**, but rather the second coupling point **42** is reliably sealed.

(63) The external thread **60** is arranged on the upper collar **50** of the base part **20** and comprises part of a bayonet catch **63**. The other part of the bayonet catch **63** is integrated into the internal thread **61** of the receptacle **40** (cf. FIG. **7**). A catch **64** is located on the upper collar **50**, which then snaps into a recess provided on the receptacle **40** when the base part **20** is screwed sufficiently deep into the receptacle **40**. The engagement of the catch **64** can be felt when it is twisted in, so that the person assembling the drain device **100** receives a haptic and/or acoustic signal that the predefined twisting depth has been reached. In addition, once the catch **64** has engaged, the base part **20** can only be unscrewed from the receptacle **40** if the catch **64** has previously been pushed out of the recess, for example by means of a screwdriver. The catch **64** thus secures the base part **20** against unintentional unscrewing.
(64) FIGS. **19** and **20** show the spacer **49** provided in the exemplary embodiment shown. The spacer **49** has a cylindrical shape and acts as a distance piece between the compression ring **44** and the sealing ring **43**. When the compression ring **44** is screwed in, the spacer **49** is inserted linearly along its central axis into the base part **20** and finally presses on the sealing ring **43**. The sealing ring **43** in turn rests on the lower collar **48** of the base part **20** and is crimped by the screwing in of the compression ring **44**, thus forming the press-fit connection **56**. In the exemplary embodiment shown, the spacer **49** further comprises a bonded sealing ring **52** which is dimensioned to form a low resistance press fit with the drain pipe **16** when assembled.
(65) The spacer **49** also has two nominal separation points **53** at which it can be easily separated into two half-shell-like individual parts, for example by means of a cutter knife. As already explained above, such a division of the spacer **49** is particularly helpful if a spacer **49** was inadvertently forgotten to be inserted into the base part **20** during assembly and, moreover, the upper inlet opening of the drain device **100** is too small to subsequently push the spacer **49** through and place it in the intended location. A spacer **49** divided into two half shells can thus also be pushed through an inlet opening that is too small for an undivided spacer **49**.
(66) FIG. **21** shows the sealing ring **43**, which is a standard sealing ring made of a flexible material with an outer diameter D**4**.
(67) FIG. **22** shows the compression ring **44** with the connecting elements **46** arranged on the upper side for producing the form-fit connection **47** to the insert **45**. Furthermore, a flattened area can be seen on the external thread **55**, which forms part of the bayonet catch **63**. The internal thread **54** of the base part **20**, which is compatible with this external thread **55**, also has a flattened area analogously. If the base part **20** and compression ring **44** are rotated with slight pressure and come into the position where the two flattened areas overlap, the bayonet catch **63** engages and thereby positions the base part **20** and compression ring **44** in relation to each other in such a way that the risk of canting—and thus: the risk of thread damage—is reduced.
(68) FIG. **23** shows the insert **45** with the connecting elements **46′**, which in the exemplary embodiment shown are U-shaped notches made in the outer contour. Furthermore, the insert **45** comprises openings **58** through which the waste water entering through the inlet opening can flow towards the drain pipe **16**. The insert **45** is therefore not only used to screw in the compression ring **44**, but rather can remain in the drain device **100** after the compression ring **44** has been screwed in. On the one hand, this ensures that the insert **45** is always there when disassembly is required, for example for maintenance work. On the other hand, the insert **45** has a sieve-like effect and reduces the risk of blockages in the drainage system.
(69) On the insert **45** there are also arrows indicating the direction of rotation for screwing the compression ring **44** into the base part **20**. In the present case, it is intended that the insert part **45**—as seen in top view—is to be turned counter-clockwise in order to make the press-fit connection **56**. However, if the insert **45** is rotated, some of the torque may be transmitted to the base part **20** under certain conditions, for example if the screw connection **38** is difficult to turn. It is therefore not impossible that the base part **20** rotates when the compression ring **44** is screwed into the base part **20**. However, the base part **20** has the further screw connection **62** by means of which it is screwed

into the receptacle **40**. In order to prevent the screw connection **62** from being loosened when the compression ring **20** is screwed in, the directions of rotation of the two screw connections **38** and **62** are designed in such a way that when the compression ring **44** is tightened, in the case where part of the torque is transmitted to the screw connection **62**, i.e., to the connection point to the receptacle **40**, this second screw connection **62** is not loosened, but rather additionally tightened.
LIST OF REFERENCES
(70) **10** floor panel **11** shower floor panel **12** opening (in **10** or **11**) **13** inlet opening (in **100**) **14** cover (for **13**) **15** extension element (for **14**) **16** drain pipe **17** external thread (of **15**) **18** snap-fit connection **19** opening (in **30**) **20** base part (of **200**) **21** bearing surface **22** inner contour (of **20**) **23** recess (in **20**) **24** opening (in **25**) **25** retainer (for **15**) **26** outer contour (of **25**) **27** upper side (of **25**) **28** underside (of **25**) **29** internal thread (of **25**) **30** fixing element **31** upper side (of **30**) **32** underside (of **30**) **33** snap ring **34** projection (on **30**) **35** recess (in **25**) **36** projection (on **30**) **37** - **38** screw connection (of **44** and **20**) **39** seal (in **20**) **40** receptacle **41** first coupling point **42** second coupling point **43** sealing ring (in **200**) **44** compression ring **45** insert (sieve) **46**, **46′** connecting element **47** connection (of **44** and **45**) **48** lower collar (on **20**) **49** spacer **50** upper collar (on **20**) **51** pipe stub **52** seal (in **49**) **53** nominal separation point (on **49**) **54** interior thread (on **20**) **55** external thread (on **44**) **56** press-fit connection **57** bayonet catch **58** openings **59** receptacle (in **45**) **60** external thread (on **20**) **61** internal thread (on **40**) **62** screw connection (of **20** and **40**) **63** bayonet catch **64** catch **65** recess (on **20**) **66** intermediate support (below **14**) **67** flange (on **25**) **68** openings (in **15**) **69** flange (on **15**) **70** length piece (on **15**) **100** drain device **150** adjusting device **200** adapter A**1** section view (from FIG. **4**) A**2** section view (from FIG. **6**) D**1** outer diameter (of **25**) D**2** inner diameter (of **40**) D**3** inner diameter (of **48**) D**4** outer diameter (of **43**) RH direction (horizontal) RV direction (vertical) SH horizontal play SV vertical play Z**1** pre-assembled state

Claims

1. An adapter (**200**) for a drain device (**100**) that is provided for installation in an opening (**12**) of a floor panel (**10**), preferably a shower floor panel (**11**), and wherein the drain device (**100**) comprises: a receptacle (**40**) for the installation in the floor panel (**10**) with an upper inlet opening (**13**) through which waste water can flow into the drain device (**100**), and parts which can be preassembled in the floor area and which can be connected in a fluidic manner to a drain pipe (**16**) and the receptacle (**40**), so that waste water which has passed through the inlet opening (**13**) can enter the drain pipe (**16**), wherein the adapter (**200**) comprises the following components: a base part (**20**) having an internal thread (**54**), wherein the base part (**20**) comprises a first coupling point (**41**) to the drain pipe (**16**) and a second coupling point (**42**) to the receptacle (**40**) and couples the receptacle (**40**) to the drain pipe (**16**) in the assembled state, a sealing ring (**43**) that can be installed in the base part (**20**) to seal the adapter (**200**) from the drain pipe (**16**), and a compression ring (**44**) with a outer thread (**55**) compatible to the inner thread (**54**) of the base part (**20**) for exerting a compressive force on the sealing ring (**43**), whereby the sealing ring (**43**) can be pressed against the drain pipe (**16**) by means of the compression ring (**44**) and the first coupling point (**41**) comprises a press-fit connection (**56**).
2. The adapter (**200**) as claimed in claim 1, further comprises comprising an insert (**45**), which can be installed in the base part (**20**) above the compression ring (**44**), for clamping the sealing ring (**43**) installed in the base part (**20**), wherein the compression ring (**44**) and the insert (**45**) preferably have mutually compatible connecting elements (**46**, **46′**) for producing a positive-fit connection (**47**), so that a torque exerted on the insert (**45**) can be transmitted to the compression ring (**44**).
3. The adapter (**200**) as claimed in claim 2, wherein the insert (**45**) comprises openings (**58**) for collecting impurities present in the waste water, for example hair, and has a sieve-like action.
4. The adapter (**200**) as claimed in claim 2, wherein the insert (**45**) has a receptacle (**59**) for a tool, in particular a screwdriver, so that the insert (**45**) can be screwed into or unscrewed from the

adapter by means of a tool.
5. The adapter (**200**) as claimed in claim 1, further comprising a spacer (**49**) provided between sealing ring (**43**) and compression ring (**44**).
6. The adapter (**200**) as claimed in claim 5, wherein the spacer (**49**) comprises a seal (**52**) for forming a guide with respect to the drain pipe (**16**).
7. The adapter (**200**) as claimed in claim 6, wherein the seal (**52**) is adhesively bonded in the spacer (**49**).
8. The adapter (**200**) as claimed in claim 5, wherein the spacer (**49**) has at least one nominal separation point (**53**) at which it can be divided.
9. The adapter (**200**) as claimed in claim 1, wherein the first coupling point (**41**) comprises a bayonet catch (**57**).
10. The adapter (**200**) as claimed in claim 1, further comprising a thread (**60**) provided on the base part (**20**) which is compatible to a thread (**61**) applied to the receptacle (**40**), so that the second coupling point (**42**) is formed by a screw connection (**63**) which couples the base part (**20**) to the receptacle (**40**).
11. The adapter (**200**) as claimed in claim 10, wherein the thread (**60**) of the base part (**20**) is a first part of a bayonet catch (**63**), which is compatible to a second part of the bayonet catch (**63**) attached to the receptacle (**40**).
12. The adapter (**200**) as claimed in claim 10, wherein the second coupling point (**42**) comprises a catch (**64**), which can be snapped into a recess compatible therewith in the receptacle (**40**) so that the base part (**200**), which can be screwed into the receptacle (**40**), can be fixed and secured against unscrewing.
13. The adapter (**200**) as claimed in claim 1, wherein the first coupling point (**41**) for connecting the adapter (**200**) to the drain pipe (**16**) and the second coupling point (**42**) for connecting the adapter (**200**) to the receptacle (**40**) each comprise a thread oriented in opposite directions, so that when one coupling point (**41**, **42**) is tightened, the other coupling point (**41**, **42**) is also tightened.